Kannan M

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PROFESSIONAL SUMMARY

Software Developer with nearly 1 year of professional experience in developing and enhancing enterprise applications using Python, React, Node.js, SQL, and API integrations. Hands-on experience in claims processing, document extraction, verification workflows, Python automation, database-driven applications, and UI development. Independently developed an AG Version Tool and a full-stack Claim Verification Tool, while contributing to enterprise application enhancements, performance improvements, testing, and deployment support.

PROJECTS

AG Claims Processing Platform — Extraction & Verification

Python | Streamlit | Flask | AWS Textract | SQL Server | PDF Processing | JavaScript

- Developed and enhanced an end-to-end claims processing workflow for Dental and Professional insurance claims, covering document extraction, database storage, verification, and file processing.
- Implemented automated document processing including file loading, PDF splitting, extraction, validation, version tracking, and SQL Server storage.
- Integrated AWS Textract as the primary OCR engine with Tesseract OCR as a fallback for claim data extraction.
- Developed the AG Version Tool independently to manage extraction versions, prevent duplicate processing, and maintain processing history.
- Implemented split-file origin mapping to maintain traceability between original claim documents and generated claim-level files.
- Enhanced the verification workflow with claim assignment, queue management, field editing, service-line updates, status tracking, and source-document review.
- Implemented automated file movement, logging, and CSV export for completed verification workflows.
- Worked on handling left-out/unprocessed files and maintaining detailed processing logs for troubleshooting and traceability.

Workflow:

Source Documents → File Loading → PDF Split → OCR Extraction → Validation → SQL Server → Verification → File Movement / Export

RAT — Dynamic Quality Measures & Enterprise Application

React | Python | SQL | API Integration

- Developed a dynamic **Quality Measures** section supporting multiple measures and services.
- Implemented a **configuration-driven approach** to reduce repeated hard-coded development when adding new measures or services.
- Worked on **Coder Dashboard** and **Coder Review** functionality, including query and performance improvements.
- Added and enhanced fields in **Member Chart** based on application requirements.
- Organized and improved **audit-log functionality** for better activity tracking.
- Worked on multiple database queries and application enhancements to improve functionality and performance.
- Handled RAT development and supported deployment-related activities.

Impact: Improved maintainability of configurable functionality and contributed to better application performance and usability.

Claim Processing, Extraction & Verification — Dental & Professional Claims

Python | Flask | Streamlit | AWS Textract | Tesseract OCR | SQL Server | JavaScript | PDF/TIFF Processing

- Developed and enhanced an end-to-end claim processing workflow for Dental and Professional claims, covering document extraction, processing, database storage, and verification.
- Worked on document extraction and processing, including file loading, PDF splitting, OCR extraction, validation, version tracking, and SQL Server storage.
- Integrated AWS Textract for OCR processing with Tesseract OCR as a fallback for document extraction.
- Implemented file-processing logic to handle source documents, split claim files, and maintain traceability between original and processed files.
- Developed verification functionality to display source documents alongside extracted claim data for review and correction.

- Implemented claim queue management, assignment, field-level editing, service-line updates, and claim status management.
- Added validation and tracking functionality to support accurate claim verification and reduce manual handling.
- Implemented automated file movement based on claim processing and verification results.
- Worked on processing logs, left-out/unprocessed file handling, and version/history tracking.

Workflow :

Source Documents → File Processing → PDF Splitting → OCR Extraction → Validation → SQL Server → Claim Verification → File Movement / Export

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, PHP, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend & Frameworks: Python, Streamlit, Laravel, Node.js, Flask, FastApi
Databases: Microsoft SQL Server, MySQL
OCR & Document Processing: Tesseract OCR, PDF Processing, Document Splitting, Data Extraction
AI & Developer Tools: GitHub Copilot, AI-assisted Code Generation, Claude, ChatGPT, Google Antigrivity
DevOps & Cloud: Git, GitHub Actions, Docker, Linux, CI/CD, AWS
Testing: Manual Testing, Functional Testing, QA Coordination

EDUCATION

Bachelor of Engineering (B.E.) – Computer Science and Engineering
Erode Sengunthar Engineering College, Tamil Nadu
2021 – 2025 | CGPA: 7.96

CERTIFICATIONS

Basics of Python – Infosys Springboard
- Successfully completed a foundational Python programming course through Infosys Springboard

Facebook Login UI Bootcamp – Devtown
- Completed an intensive 7-day bootcamp focused on developing a Facebook login page using **HTML & CSS**.

The Ultimate DevOps Bootcamp – In Progress
- Currently enrolled in a comprehensive DevOps bootcamp covering **CI/CD, Docker, Kubernetes, AWS, GitHub Actions**, and other industry tools.

ACHIEVEMENTS

Paper Presentation – 1st Place
Sona College of Engineering and Technology
- Secured First Position in a technical paper presentation on Automatic Number Plate Recognition.

CO-PO Attainment Certificate
Erode Sengunthar Engineering College
- Developed a software tool to track Course Outcome–Program Outcome (CO–PO) attainment. Recognized for this contribution during the 2024–2025 academic year

PROFESSIONAL PROFILES

- **Portfolio :** https://kannan-24.netlify.app
- **GitHub :** https://github.com/Kannan-24
- **LinkedIn :** https://linkedin.com/in/m-kannan